

SINO-GERMAN WORKSHOP ON CHEMICAL AND PHYSICAL INTERACTIONS BETWEEN PARTICLES AND FLUIDS

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1. Introduction

Supported by the Sino-German Center for Research Promotion, and organized jointly by the National Natural Science Foundation of China (NSFC) and the German Research Foundation (DFG), the third Sino-German workshop on particle fluid systems was held on Oct. 24-31, 2004 in Beijing, China, following the two previous successful workshops held on Aug. 30-31, 1999 in Hamburg, Germany (Li and Werther, Chem. Eng. Technol., 23(4), 378, 2000) and May 18-19, 2001 in Beijing, China (Li, Ge, Werther and Bruhns, Chem. Eng. Technol., 24(11), 1097, 2001). Thirty-one scientists from China, Germany, Japan and The Netherlands came together for interdisciplinary discussion over the core problem of multi-phase reaction systems in the name of "Chemical and Physical Interactions between Particles and Fluids".

Compared to single-phase systems, multi-phase systems are more commonly encountered in engineering, and they possess unique complexities due to the discontinuities and abrupt changes on the phase-interfaces. In particular, the sharp difference between solids and fluids in mechanical and physical properties has brought about a gallery of complex phenomena involving the physical and chemical interactions between the two media. However, it also brings about various possibilities for developing new techniques and processes for high-efficiency and well-controlled transformation of mass and energy that can improve the quality, productivity and ecological economy of the huge amount of products in process industries. As a result, particle-fluid and particle-particle interactions (which naturally happen in fluids) come to the focus of scientists and engineers in all process industries. The workshop covers some of the research areas ranging from fundamentals to applications, which have been divided into six sections, namely, the particle-particle and particle-wall interactions, fine particle fluidization and related fluid-particle interactions, modeling and simulation of particle-fluid interaction both from the micro-scale, and from the meso and macro scale, particle engineering as well as some related industrial applications.

This report follows the structure of the workshop and usher us to revisit the six sections, each based on the review contents by courtesy of some participants, which are followed by a conclusion section at the end.

2. Session Reports

2.1 Particle-particle and particle-wall interaction

Both experimental and modeling studies were reported in this session to quantify particle-particle and particle-wall interactions. A thorough understanding of these phenomena is vital to a wide variety of processes involving particulate solids. During the session five experts presented their recent work which will subsequently be briefly summarized together with possible directions for future research indicated during the discussions.

Prof. Ripperger (Technical University of Kaiserslautern) reported a detailed experimental study on the measurement of the normal adhesion force to walls. Pull-off force measurements with an Atomic Force Microscopy (AFM) method and a vibration method showed good agreement for small particle sizes ($d_p < 10 \mu\text{m}$) for various particle-wall combinations. However, a dominating influence of micro asperities on the measured adhesion force for very rough substrates was reported. Further study is required to control and quantify the effect of adsorbed species and surface roughness. Extension of this promising technique to larger non-spherical particles seems desirable and could contribute significantly to our understanding of particle-wall interaction.

Prof. Schmidt (University of Wuppertal) expanded in his talk on the measurement of detachment forces of differently structured dust layers from surfaces under well-defined conditions. This work is directly relevant for the prediction of the force required for particle removal from walls in process equipment. However, the extrapolation of these interesting findings to industrial conditions requires substantial additional efforts. In this connection it would for instance be interesting to study in more detail the impact of the mode of particle layer creation on the detachment force. Moreover it would be interesting to assess to what extent the detachment force can be predicted on the basis of detailed modeling.

Prof. Peukert (Friedrich-Alexander University of Erlangen-Nuremberg) contributed with a stimulating talk focusing on the understanding of particle-particle interactions from a microscopic view of particulate interfaces. Various methods were discussed to measure the adhesion forces together with modeling and simulation approaches. Par-

ticularly interesting were the results for particle-surface contact obtained from Molecular Dynamics (MD) simulations revealing the formation of nano-wires upon separation. It would be very interesting to establish whether nano-wire formation constitutes a generic major mechanism upon particle-particle separation. Also more work is required to assess the role of humidity during particle-particle interaction. Clearly the incorporation of this type of knowledge in macroscopic (engineering) models is clearly required and could significantly boost the predictive capabilities of these models.

Prof. Horio (University of Agriculture and Technology of Tokyo) addressed the incorporation of lubrication forces in Discrete Element Simulation (DEM) techniques. Traditionally these forces are neglected in DEM simulations of gas-particle systems but especially for fine particles the incorporation of lubrication forces in the equation of motion for the particles seems important. The availability of a simple criterion, to assess the necessity to account for lubrication forces, seems desirable. In addition it would be interesting to investigate the impact of lubrication forces on the macroscopic flow structure in dense gas-fluidized beds.

Prof. Schmidt (University of Wuppertal) presented in his second talk a methodology to compute from first principles the magnitude of the particle-wall adhesion force for certain idealized particle surfaces. Although a number of assumptions had to be made, which require verification, the methodology seems very promising to compute the particle-wall adhesion force in a fundamental way provided that surface roughness can be accounted for in a reliable way. The extension of the methodology to different particle (surface) geometries is also desirable.

Prof. Li (Tianjin University) reported on a detailed experimental study to accurately describe the mechanical reliability of solid catalyst. A number of existing methods were critically reviewed also. This work constitutes an important first step to arrive at improved procedures for assessing the reliability of catalysts from a mechanical point of view. Ultimately a methodology will be developed including not only mechanical but also thermal and chemical factors which should then be incorporated in design procedures of Packed Bed Reactors (PBR).

2.2 Fine particle fluidization and particle-fluid interaction in fluidized bed systems

The major challenge in fluidisation of fine particles is the understanding of powder cohesion, i.e. of particle interactions relative to the interactions of the fluid with clusters, aggregates and primary particles in the fluidised bed and related apparatus such as riser or downer reactors as well as cyclones. For the understanding, design and control of fluidisation of cohesive powders clearly a multi-scale approach is needed which covers several length and time scales between particle-particle interactions which are inherently controlled on the nanoscale and large scale technical systems.

Prof. Jin (Tsinghua University) pointed out how bubbles, walls, agglomeration may influence the performance of various types of fluidised beds positively as well as negatively. Prof. H. Li (Institute of Process Engineering) stressed that in a fluidised bed various regions with different types of aggregates may exist, and he distinguished between agglomerate particulate fluidisation (APF) with no bubble formation and agglomerate bubbling fluidisation (ABF). The latter may be better controlled through tailoring the adhesion forces between the particles so that bubble formation can be partly suppressed.

An especially interesting case was presented by Prof. Wei (Tsinghua University) for catalysed carbon nanotube production in fluidised beds. The carbon nanotubes form loose structures which can be easily fluidised although the bed expansion may exhibit hysteresis. One fascinating aspect in fluidisation is that new as well as traditional applications allow great progress. Prof. Chen (China University of Mining and Technology) presented an overview on the state-of-the-art of a dry Clean Coal Technology which will be very important for arid regions in China. He did show how segregation between coal and gangue due to density can be achieved in an air-dense fluidised bed (e.g., magnetite powder), which has led to higher recovery in coal mines, improved combustion in power plants, and cost reduction for flue gas desulfurisation. Fine particles are, however, also a challenge in related apparatus such as cyclones which have to be operated often under extreme conditions such as high temperatures and pressures. This aspect was dealt with by Prof. Sun (University of Petroleum) who pointed out that the pressure drop in cyclones drops due to decreased fluid density at high temperatures, whereas the separation efficiency decreases considerably since the inertia-dominated separation depends on fluid viscosity which increases with increasing temperature.

In summary, while the general trend towards finer and finer particles strongly influences fluidization, it poses many interesting problems for research and calls for co-operation on both theoretical and experimental sides.

2.3 Modeling and simulation of particle-fluid interaction — the micro scale

This session consisted of five presentations with three from the Chinese side and two from the German side. Prof. Gu (East China Normal University) reported on the simulation of the charge relaxation process in particle-fluid systems. The dynamic electrorheological effects of the rotating particles was analytically formulated and the differential equations for the Quincke rotation were established, which explained the breakdown of the symmetry for the rotation axes in the suspension.

Prof. Xu (China Agricultural University) introduced their research on particle-particle interactions in pure granular flow or particle-fluid systems. The discrete element method were applied to simulations on some typical processes for dry particles, such as particle piling, silo discharge, braking

and vibrating of a truck-box etc. A model for both normal and tangential interactions between two spheres with an interstitial power-law fluid was reported, which led to analytical formulation for the viscous force components for either shear-thinning or shear-thickening fluids. The effects of partial slip on the normal interaction were analysed.

Dr. Ge (Institute of Process Engineering) tried to advocate that particle methods should be ideal constituents and powerful tools of multi-scale models, owing to their physical fidelity and computational simplicity. According to him, different particle methods may be suitable for simulations on different scale for granular flows and particle-fluid systems, and the so-called pseudo-particle modelling they have been developing may present a promising choice for describing the dynamic transport process on the micro-scale in heterogeneous particle-fluid systems.

Prof. Nirschl (University of Karlsruhe) demonstrated in his presentation the ability of the moving particle-particle in cell (MP-PIC) method in simulating particle interactions at the micro-scale using overlapping grids. It was argued that such a treatment may be necessary in many chemical engineering processes where the particle extension needs to be considered, i.e. to calculate mass transfer in particle coating processes or fluid stresses for dispersing particles in liquids.

And Dr. Kruis (University Duisburg-Essen) discussed the drag characteristics of sub-micron nano-particle aggregates and their calculation. A newly developed Direct Simulation Monte Carlo (DSMC) method was employed for the work, which allowed him to trace changes in complex multidimensional systems, in this case describing particle size, charge and aggregate composition. The study was focused on the aggregate composition in dependency on initial size and charge distribution.

Though this session addresses in detail some of the problems and results related to particle/fluid interactions on the micro scale, similar problems were presented in other sessions, which are included in the following summary.

From the viewpoint of an individual particle, the micro-scale interaction may be related to different aspects, namely, particle-fluid, particle-field (electric, magnetic, gravity, ...), particle-particle and particle/wall. There have been numerous approaches to consider these interactions. These are based on numerical modelling and simulation such as molecular dynamics (MD), lattice-Boltzmann, Lagrangian tracking, two-fluid and multi-scale methods, or experimental methods. Even analytical analysis and methods can deliver useful basic information on fluid/particle interactions as has been proved in some contributions of this session.

Most of the contributions looked at drag formation on individual particles in a complex environment. The major achievements that may be drawn from the papers are included in some new models for chemical engineering applications (in terms of correlations and/or phenomenological approaches and detailed analysis), though typically they are not mature enough to be transferred into general

simulations for application, that is, parameterization of the results may be still too complex to allow transfer.

The open question that may be derived from the conclusions of those papers is that experimental or otherwise verification of the micro-model is yet a future challenge which mainly involves the transfer and reflection between scales, that is, the transfer of models to the next upper scales, into the meso-scale or directly into the macro-scale. It has been pointed out also that a new scale below the micro-scale (based on the characteristic length of a single particle) has to be addressed, namely a submicro-scale, for which surface properties as well as e.g. surface roughness effects and/or chemical interactions need to be investigated.

The session has raised some typical points of possible cooperation, e.g., interactive investigations are to be seen in between the individual scales of modelling fluid-particle interactions, e.g., the micro- to meso- to macro-scale. An example of such an approach is included in the statement of Prof. Qi (Tsinghua University, Beijing, China). Also an approach where individual effects on the micro-scale should be addressed by different analysis methods is worth to consider for future investigations, for which once again the model verification plays an important role. Finally, a major area of investigation that still needs attention and future work is particle-turbulence interaction on all scales, the micro-, meso- and macro-scale.

2.4 Modeling and simulation of particle-fluid interaction — the meso and macro scale

This session had five contributions which can be divided in principle into two categories. The first three presentations described the flow physics in a circulating fluidised bed with classical physical methods of fluid simulation techniques. The first presentation by Prof. J. Li (Institute of Process Engineering) was a multi-scale analysis of particle-laden flow in a circulating fluidised bed on the basis of the EMMS approach. The governing equations for the dense and dilute phases lead to an insufficiency of the conservation conditions. A compromise has been found between the dominant mechanisms, that of the particles and of the fluid. The solution lies in the physical concept of the EMMS (Energy-Minimization Multi-Scale) model with a multi-scale description of an inter-scale correlation by analysing the compromise between the dominant mechanisms. The pseudo-particle model provides the verification of the stability condition in a micro-scale description for the analysis of macro-scale phenomena.

In the second presentation by Prof. Kuipers (Twente University) pointed out the multi-scale modelling. He stated that the development of a single universal model is still far too ambitious. It seems that the flow could be described by modelling dense gas solid flows with four levels of scaling. On the highest scale, a discrete bubble model is appropriate where individual bubbles are tracked in the computational domain by taking into account bubble-bubble and bubble-wall encounters. On the next smaller scale, his

choice consists of a continuum model based on kinetic theory as a statistical mechanical description of particle-particle encounters. It is based on a more fundamental description of particle-particle interaction compared to the classical two-fluid models and the discrete bubble model. With decreasing scale we come to the discrete particle model where individual particles are tracked in the computational domain by taking into account particle-particle and particle-wall collisions. Here the number of particles is already very limited but still high compared to the lowest Lattice-Boltzmann model. Fictitious particles propagate in a three-dimensional lattice and collide at discrete positions. All details of the flow field can be obtained. Critical for each level are the closure conditions for gas-particle and particle-particle interaction. Related to the flow problem it has to be worked in multi-scale approach as long we are not able to resolve all the details by MD-simulations.

Dr. Yang (Institute of Process Engineering) came back to the EMMS strategy described in the first contribution. Once again, he pointed out the possibility of a multi-scale resolution of structure and gas-solid interaction in the EMMS model. He implemented the EMMS strategy in a CFD package and showed the dynamic flow structures for a circulating fluidised bed. The drag coefficient is found to be an important factor for two-fluid models, and the EMMS model is an improvement for the prediction of meso-scale structures and solid entrainment rate. In summary, the multi-scale approaches have undergone significant improvement. Though it is still not possible to abandon experiments, the simulations have shed new light on the physical behaviour of the multiphase flow.

The second categories used a global 'black box' approach without trying to describe the flow physics in a fluidized bed reactor. Prof. Shi (Sichuan University) pointed out a multi-fractal analysis of the particle fluid system where a wavelet transform modulus maxima method was used to analyse the fluctuation signals. The singularity spectrum shows that the gas-solid flow has the multi-fractal character and that the motion behaviour can be described by such a spectrum. The intermittency index, which describes the non-uniformity of the flow pattern, can be used to quantitatively determine the transition of flow regimes from fast fluidization to pneumatic conveying.

An artificial neural network approach (ANN) to the prediction of the nonlinear dynamics was presented by Dr. Tsutsumi (University of Tokyo). It could be shown that the trained ANN provides accurate prediction of local voidage fluctuations. The training data have been acquired for a CFB riser installed with pressure transducers, optical transmittance probes and heat transfer probes. The author investigated a single and long-term prediction where the effectiveness of the early-stop training approach has been confirmed. A well-trained ANN captures certain nonlinear dynamic properties of the real system which were shown in comparison to experimental results in a very impressive way.

Both categories of papers are very helpful for the under-

standing of the flow physics and in the control of CFBs. Although we have reached a very high level, there still remain challenges in the future. The simulation techniques call for improvement and anticipate merging of models between different scales. The 'black box' approaches have to show their usefulness by correlation with physical behaviour especially in conditions beyond available training data or generalized flow structures. The largest challenge is to bring the two approaches together and correlate global signals with different physical models. Besides, the product and particle properties should be taken into account. In general most of the investigations rely on standard particles which can easily be handled. General results should be transferred to particle properties of industrial materials which are difficult to describe. When we have truly succeeded in this integration we are ready to replace expensive experiments by simulation methods for industrial applications.

2.5 Particle engineering: mixing, handling, spraying technologies and macro scale properties

This session was divided into three sub-sessions: "Mixing and Handling of Fine Powders", "Engineering of Macroscale Properties", and "Particle Engineering by Spraying Technologies". The papers presented had the titles: "Electrostatically Supported Mixing of Fine Grained Particles" (Prof. Wirth, University of Erlangen-Nuremberg), "Brownian Coagulation of Charged Particles" (Dr. Kruis, University of Duisburg-Essen), "Particle Coating by Chemical Vapour Deposition in a Fluidised Bed Reactor" (Prof. Werther, Technical University Hamburg-Harburg), "Particle Coating in the Supercritical Fluidised Bed" (Dr. Hartge, Technical University Hamburg-Harburg), and "Droplets and Particles in Sprays" (Prof. Fritsching, University of Bremen).

New ideas and significant achievements in general can be summarized as follows: Tailor-made particle properties can be obtained by a variety of different techniques like spraying, coating, aggregation, and agglomeration; the latter can be carried out very selectively by using electrostatic forces. The coating of solid particles can take place in the gas phase, liquid phase, and even supercritical phase. The coating material can be primarily solid, liquid, gas, or solution. The materials used can be electrically charged, can coagulate, can change its phase (phases), and can react. By selecting the proper techniques for given materials the desired particle properties can be acquired. Related fundamentals of multiphase flow help significantly to understand the processes involved.

Special particle properties that can be generated or adjusted by the above mentioned techniques cover a wide range, e. g., the flowability and agglomerate size of lactose particles could be changed significantly by electrostatically enhanced coating with nano-particles. This is relevant for powder inhalers for the administration of pharmaceutical substances. Another application of these functionalized micro-particles is in thin paint coating. The Brownian coagulation of charged nano-particles, like tin-dioxide and

silver, finds application for gas sensors, to result in higher sensitivity and more rapid response. Besides coagulation, an even thinner and more homogenous film, e.g. on glass beads can be achieved by thermal decomposition of tri-isobutyl-aluminum, via the so-called chemical vapour deposition (CVD), or by wetting with a paraffin solution in supercritical carbon dioxide. These substances have been chosen as quasi-standard materials to get input data for modelling and simulation codes that are developed in parallel. The usage of somehow similar software-packages was shown when describing the behaviour of special sprays derived from viscous ceramic melts.

Open points and matters of dispute are given by the following question: Which technique is suitable for the desired product properties (in terms of quality, costs, sustainability ...)? What are the restrictions of the different techniques (in terms of mass throughput, educt requirements ...)? Which features of models or simulation procedures are necessary (e.g. related to temperature change)? How deep has one to enter the micro-scale (or even nano-scale/chemistry) to understand the meso-scale and to handle the macro-scale? It is obvious that the answers might differ for scientists and for engineers. Areas of co-operation are, for example, related to more knowledge of the processes, better control of the processes, and scale-up or scale-down of the processes.

2.6 Industrial applications of multiphase flow systems

This topic includes two sessions with 7 contributions discussing the topic from different areas of application and from different angles.

Prof. Peukert (Friedrich-Alexander University Erlangen-Nuremberg) talked about product engineering of particulate solids. Structure formation is the key concept for product engineering. Transport mechanisms (like diffusive and convective transport) in combination with interactions between the particles (like van-der-Waals- and electrostatic interactions) generate structures like those of crystals, structure of filter cakes and of particulate thin films. Small changes on the solid surfaces (e.g. ligands) may have strong influences on neighboring interactions and finally on structures which result in changed macroscopic properties like viscosity of suspensions.

Prof. Chen (Beijing University of Chemical Technology) gave a presentation on rotating packed bed as multiphase reactor for massive synthesis of nanoparticles. To generate nanoparticles via reactive precipitation two time scales have to be considered — the micromixing characteristic time and the nucleation characteristic time. When the mixing time is small in comparison to the nucleation time a narrow particle size distribution is generated. The high-gravity (some hundred) rotation packed bed reactor permits the adjustment of mixing time to values smaller than the nucleation time. Moreover, high throughputs can be achieved with this reactor.

Prof. Sun (University of Petroleum in Beijing) discussed

the separation efficiency of a PVTM cyclone. It was stated that separation efficiency of industrial cyclones at high temperatures are lower than calculated values, and a dimensionless experimental correlation of cyclone efficiency, including the influence of operating temperature was obtained. As a consequence new designs of cyclones running at elevated temperatures are necessary, which require further knowledge on particle-fluid flow and more attention to boundary layer flow, particle movement due to turbulence and particle-fluid interaction within the cyclone.

Prof. Schmidt (University of Wuppertal) gave a presentation entitled "Methods for dustiness estimation of industrial powders". Methods published in the literature and their further developments were reviewed and related particle sizing and counting techniques were also presented. Integrated measurement methods were then summarized systematically. Different mechanisms are responsible for the generation of dust, so the presentation led to interesting questions such as: Which are the relevant mechanisms? Are there similarities among the mechanisms which cause attrition? How are the processes used correlated to the mechanisms? Besides these basic tasks it is necessary to consider the measurement methods for the determination of the dustiness of powders. Are different testers needed for the different mechanisms creating dustiness?

Prof. Fang (Zhejiang University) presented their work on biomass flash pyrolysis in fluidized bed. Fluidized beds can be used for biomass flash pyrolysis to produce bio-oil. Because of the high oxygen content bio-oil has poor long-term stability. Future work is necessary to upgrade the bio-oil to use it as an added-value product. However, the costs for the bio-oil production should be considered for the process development. What are the costs for other processes using biomass (e.g. combustion)?

Prof. Qi (Tsinghua University) talked on the topic of interaction between SO₂ in flue gas and sorbent particles in dry FGD processes. For flue gas desulfurization (FGD) in coal combustion a low cost system is necessary. The circulating fluidized bed FGD can fulfil the requirements, though the pressure drop of CFB is too high to make the process economic. The precalculation of the circulating fluidized bed, using CFD and EMMS, requires detailed knowledge of the cluster drag coefficient which still lacks reliable correlations. Further work should be done on this topic. On the other hand the question is whether there are other processes appropriate to achieve similar desulfurization results with lower costs.

Dr. Yang (Tsinghua University) also gave a talk related to coal combustion concerning the combustion efficiency of coal char in CFB boilers. The efficiency can be characterized by the carbon content in the fly ash which depends on the coal used and on the process parameters like secondary air distribution, mixing of coal in the combustor and the cyclone efficiency. The secondary air distribution is very critical to the carbon content. To reduce the carbon content the secondary air has to be distributed more effectively. Further questions can be raised as to how can such

an effective air distribution be accomplished and what parameters influence the penetration of the secondary air jets.

Despite the diversity of the applications of multi-phase systems involving particles, common interest has been expressed on the quantification of the collective behavior of the particle phase, which is increasingly needed to improve or to evaluate the competitiveness of a relevant technology. Inter-particle and particle-fluid interactions are, of course, the major factors to shape such behaviors.

3. Conclusions — Research Needs and Possibilities of Cooperation

In the final discussion, which was also joined by Prof. Kwauk (Institute of Process Engineering) and Prof. Reh (Swiss Federal Institute of Technology), the participants came to the following general conclusions and prospects on the theme of the workshop:

- Firstly, it has become a common awareness that the research field of particle-fluid systems is undergoing a profound expansion from traditional chemical engineering to the fields of material science and nano-technology. The trend has been driven by technological and engineering developments and has posed new challenges to experimental, theoretical and numerical researches on these systems which require new methodologies and transdisciplinary cooperation.

- Secondly, multi-scale methods are attracting more and more attentions as an effective approach to cope with the vast scale span involved in the interactions between particles and fluids. And the key issue for this approach is to correlate the interactions on different scales. Traditional statistical approaches seem to be insufficient for this correlation and physical analysis of the underlying mechanism of dynamic structures on different scale is required for a proper correlation.
- Thirdly, the complexity of the interactions between particles and fluids has provided simulation research with great opportunities in quantifying particle-fluid systems, and at the same time, it is crucial for the development of simulation approaches to reflect real physics of the systems and to be comprehensively validated in experiments. Special experiments with strictly controlled conditions are desirable for this purpose.

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